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Chapter 7

Cell Cycle

Learning Objectives

The learner will be able to,

- *Outline the cell cycle and different stages in cell division.*
- *Recognise the importance of mitosis in the production of genetically identical cells.*
- *Have an insight on the significant of mitosis and meiosis.*
- *Understand how a single cell divides to a whole organism.*
- *Familiarize the behaviour of chromosomes in plants and animal cells during meiosis.*
- *Know about crossing over and random assortment of homologous chromosomes and its importance.*

Chapter Outline

7.1 History of cell division

7.2 Cell cycle

7.3 Cell Division

7.4 Difference between Mitosis and Meiosis

7.5 Mitogens

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One of the most important features of the living cells is their power to grow and divide. New cells are formed by the division of pre-existing cells. Cells increase in number by cell division. The parent cell divides and passes on genetic material to the daughter cells.



Edouard Van Beneden, a Belgian cytologist, embryologist and marine biologist. He was Professor of Zoology at the University of Liège. He contributed to cytogenetics by his works on the roundworm *Ascaris*. In his work he discovered how chromosomes organized meiosis (the production of gametes).

DO YOU KNOW?

Neurons can be replaced!

Stem cells in the human brain - most neurons are in G_0 and do not divide. As neurons and neuroglia die or injured they are replaced by neural stem cells

